Omar Elarabi

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EDUCATION

Queen Mary University of London, Aerospace Engineering, 3rd Year (on track 1st class) **September 2023 - Present**

- ***Key Modules Completed:*** Designing for Sustainable Manufacture, Control System Analysis and Design, Conceptual Design, Materials Engineering, Aircraft Propulsion Systems & Structures, Aerothermodynamics of Flows, Simulation Tools.

PROFESSIONAL EXPERIENCE

Peer-Assisted Study Support (PASS) Student Organiser **August 2025 - Present**

- Monitored session schedules and room allocations using Excel-based analysis, checklists of essential requirements, and coordination with departments, whilst planning promotional activities and coordinating 10 mentors to deliver sessions.
- Maintained module page and feedback analytics, liaised with staff and professors to align sessions with student needs, and implemented improvements based on data-driven insights, increasing engagement by 40%.

Peer-Assisted Study Support (PASS) Mentor **September 2024 - Present**

- Recognised and addressed specific areas where students struggled, developing strategies to address these weaknesses and improve overall performance by an average of 60%.
- Customised teaching approaches and managed sessions based on individual student needs, fostering a supportive learning environment and driving academic progress. Awarded Mentor of the Year.

ACADEMIC PROJECTS

'Designing the Powertrain Exhaust System – Formula Student car' – SolidWorks & ANSYS Fluent **September 2025 - May 2026**

- Research sustainable materials and manufacturing processes to optimise performance, weight, and sustainability. Design headers, collector, tailpipe, and muffler in CAD, ensuring fit and compliance with chassis and IMechE rules. Use CFD and thermal simulations to refine gas flow, scavenging, and heat management. Conduct lifecycle analysis to guide material selection and sustainable manufacturing. Develop and test prototypes to validate performance and noise compliance.

'Water Tank Level Control' – MATLAB & Simulink **March 2025 - April 2025**

- Modelled and tuned open- and closed-loop water tank systems in Simulink, optimising PID parameters to improve response metrics (rise time, overshoot, settling time). Processed 4,597 experimental data points in MATLAB, quantified deviations between physical and simulated systems (≤12.2% error), and demonstrated the impact of calibration choices on system stability and performance in a first-class report.

'Redesigning the A320 Flap Track Fairing – National Composites Centre' – CAD & ANSYS **February 2025 - March 2025**

- Screened over 4,000 materials in ANSYS EduPack and applied Ashby's method to select a composite for the A320 flap track fairing, meeting constraints of 3.4 kg mass, 1.66 mm deflection, and stability at 44,000 ft. Refined the material composition to maximise stiffness-to-density ratio and sustainability, defining prepreg layup and autoclave curing methods. Incorporated sustainable strategies such as downcycling analysis and precision cutting to minimise waste. Presented to NCC stakeholders.

EXTRA-CURRICULAR & LEADERSHIP ROLES

Aurora Initiative, Health & Safety Briefings Officer, QMUL **August 2025 - Present**

- Conducted safety inductions and pre-activity briefings for society members, developed accessible briefing materials, and led annual risk assessments to mitigate hazards across all events and engineering activities.
- Maintained compliance and attendance records, coordinated with other officers and departments to update safety protocols, and implemented preventative measures that ensured activities ran safely and in line with engineering standards.

Hyperlink, Braking Team Member, Queen Mary University of London **September 2024 - May 2025**

- Designed concepts using sketching and CAD, of magnetic braking methods for primary and secondary braking systems including use of eddy currents for magnetic damping and braking pads for frictional drag.
- Incorporated dimensions and material research into maximising efficiency for primary system, and methods to primary system, whilst considering health and safety, including power faults.

Second Year Aerospace Engineering Course Representative, QMUL **September 2024 - May 2025**

- Built a communication channel between a sum of 190 students and staff to improve student experience in partnership with QMUL, students and student unions.
- Raised awareness of issues in meetings, reporting feedback from meetings and recording turnouts after improvements.

SKILLS & INTERESTS

IT Skills: Computer-aided Design (CAD), MATLAB, Python, Abaqus, Microsoft Office, Computational Fluid Dynamics (CFD), ANSYS.

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